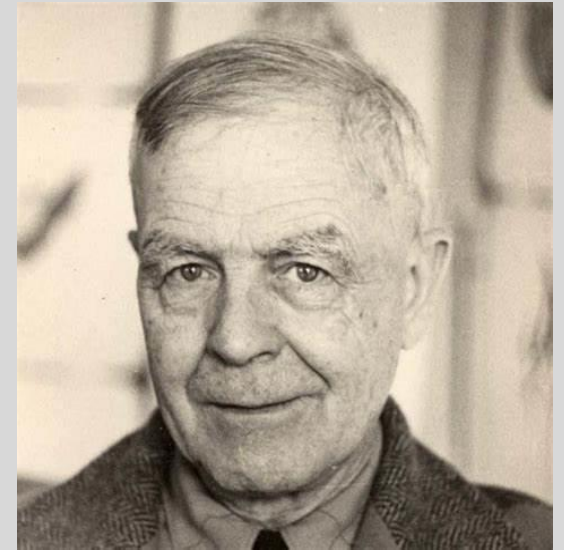


PPOM II: OMM Lab #34: OCF: Diagnosis and Treatment with Balanced Membranous Tension, V-Spread & Directing the Tide, CV4, Unilateral Temporal Hold



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William Garner Sutherland **Do.**
Make.
Heal.
Innovate.
Reinvent the Future.

Questions

Please note there were a few study questions posted prior to lab
in the CPG

1. What structures pass through the jugular foramen?

- a. Cranial nerves VIII, IX, and X and the internal jugular vein**
- b. Cranial nerves IX, X, and XI and the internal jugular vein**
- c. Cranial nerves X and XI and the internal jugular vein**
- d. Cranial nerves IX and X and the internal jugular vein**
- e. Cranial nerves VIII and IX and the internal jugular vein**

2. You diagnose a patient with a strain pattern and decide to treat them with indirect balanced membranous tension. You examine the patient with a vault contact and treat them by exaggerating the freedoms by guiding the bony contacts of all fingers slightly cephalad while bringing your fingers together slightly on either side of the head. What cranial strain pattern are you treating?

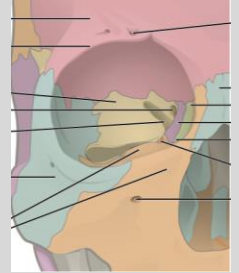
- a. Extension**
- b. Flexion**
- c. Left torsion**
- d. Right torsion**
- e. Superior vertical shear**
- f. Inferior vertical shear**

Objectives Students will:

1. Review Landmarks. Assess for restrictions at various sutures and assess the PRM. **Motion test** for cranial strain patterns to decipher **strain patterns present** in our partners (flexion, extension, torsions, sidebending-rotations, vertical shears (superior or inferior), lateral strains, and/or SBS compression). Also learn to utilize **Unilateral Temporal Contact/Hold** to help Dx temporal dysfunctions.
2. Use **balanced membranous tension** (BMT) to bring the cranial mechanism back to a more physiologic state. Practice **Indirect BMT and/or Direct BMT** on your partners utilizing one or more cranial contacts (holds).
3. Assess the Occipito-Mastoid (OM) Suture for restriction. Importance? Consider temporal dysfunction, as well as cranial nerves IX, X, XI and jugular vein that exit near this region at the jugular foramen. Learn to treat this important area with **Directing the Tide and V-Spread of the OM Suture(s)**.
4. Learn about the wide-ranging utility of the **Compression of the 4th Ventricle (CV4)** technique, how to perform it, and how it relates to enhancing the motion and amplitude of the cranial rhythmic impulse, as well as its ability to augment the patient's healing capabilities. What might be the effect of CSF stasis on Circumventricular organs and CNS function? – consider therapeutic benefit of CV4 impact on basic physiologic centers including the brainstem, etc.
5. Review material from OMM lectures and labs 32 (Dr. Ventimiglia) and 33 (Dr. Terzella). Students will strive to diagnose and palpate various cranial strain patterns as well as demonstrate OA & condylar decompression and the various lift techniques from antecedent labs.

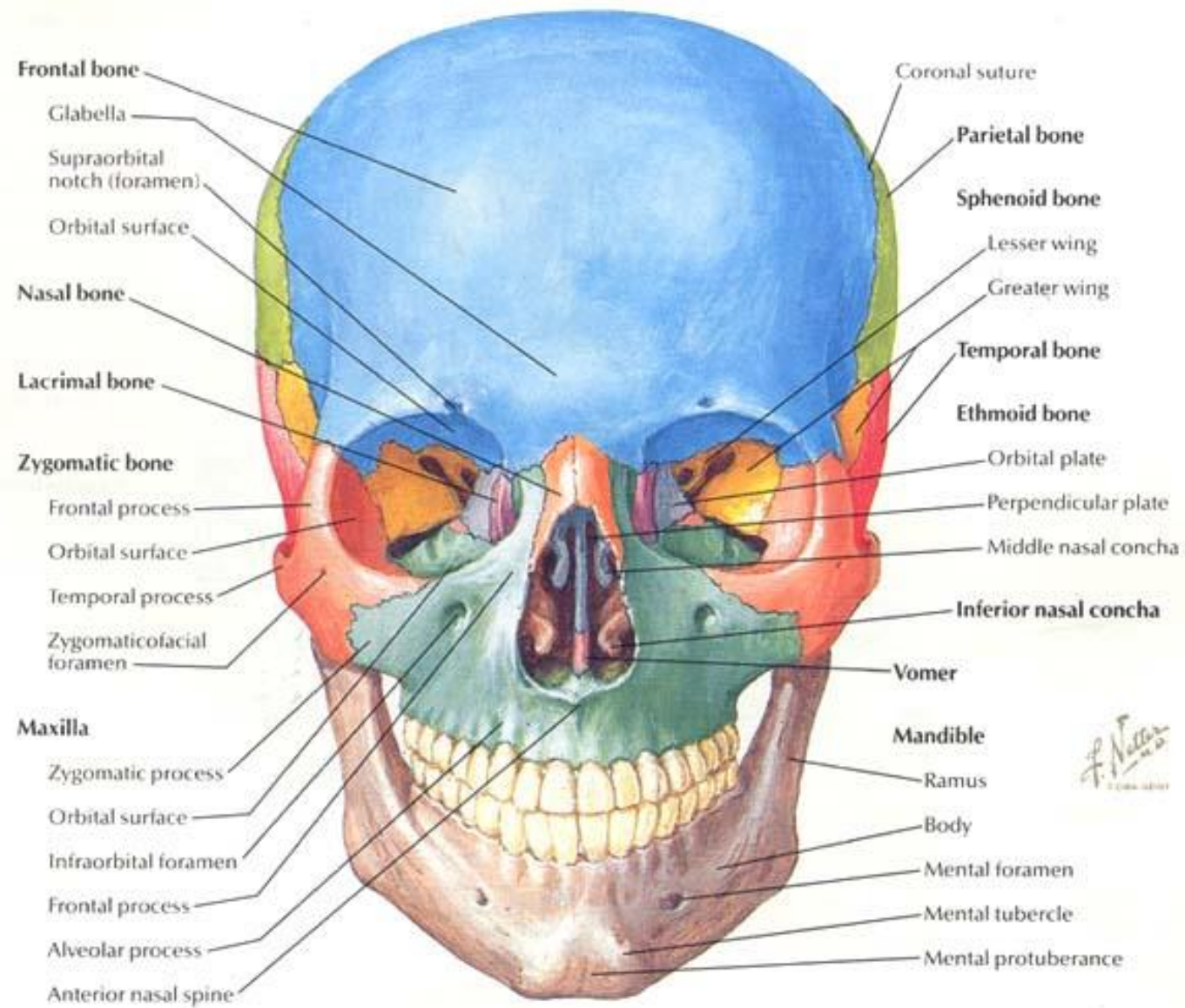
Students will continue to appreciate the complex structure and function of skeletal, arthrodial, myofascial, vascular, lymphatic, and CNS elements as a whole in the context of dxing & treating with OCF/ OCMM- osteopathic cranial manipulative medicine.

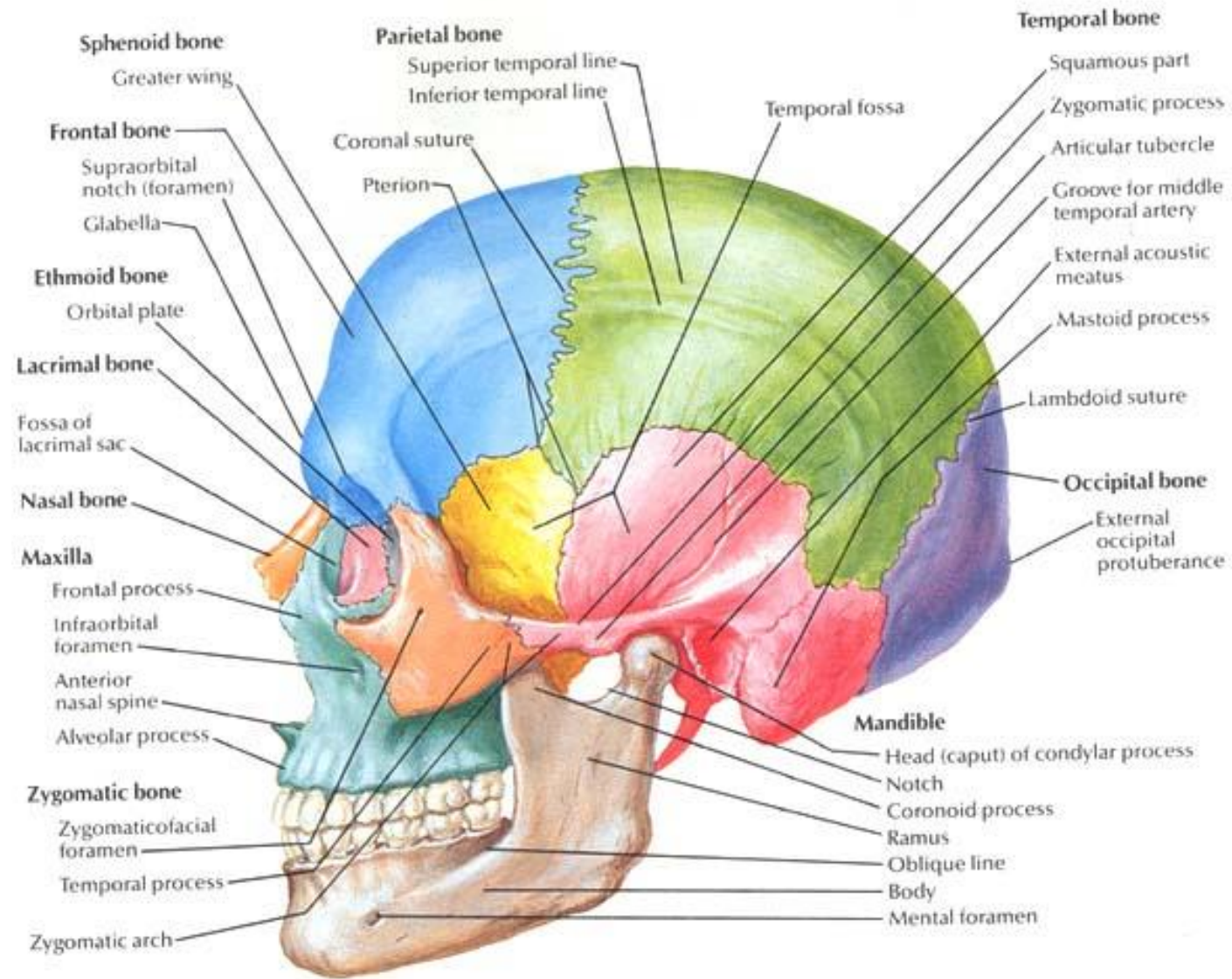
Review Landmarks!

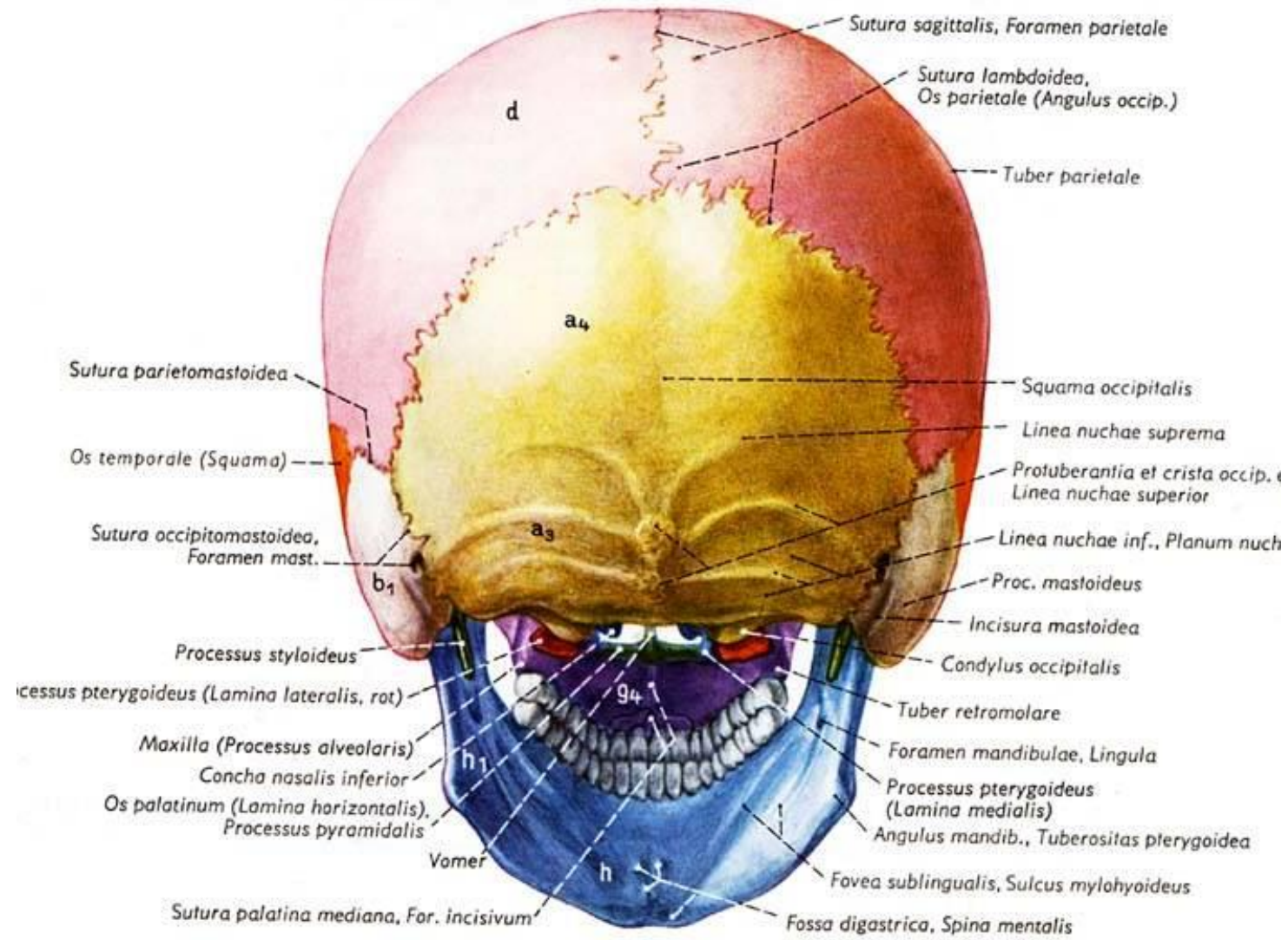


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- **Nasion**
- **Nasal(s)**
- **Frontal(s):** Zygomatic process of the frontal, Glabella, Bregma
- **Zygoma(e):** temporal process, frontal process
- **Maxilla(e):** zyg- max suture, frontal process
- **Parietal(s):** Sutures (coronal, sagittal, squamous, lambdoid)
 - transverse sinus-**Asterion**—area of most amplitude in vault contact – think venous drainage
- **Sphenoid:** greater wings, lesser wings
- **Temporal(s):** zygomatic process, squama, mastoid process
- **Occiput:** inion, lambda, nuchal lines, confluence of sinuses
- **Sutures:** coronal, sagittal, lambdoid, squamous, frontonasal, zygomatic arch, occipito- mastoid (O-M), frontal-lesser wing of sphenoid.
- **(Former) Fontanelles:** at Pterion, Asterion, Bregma and Lambda





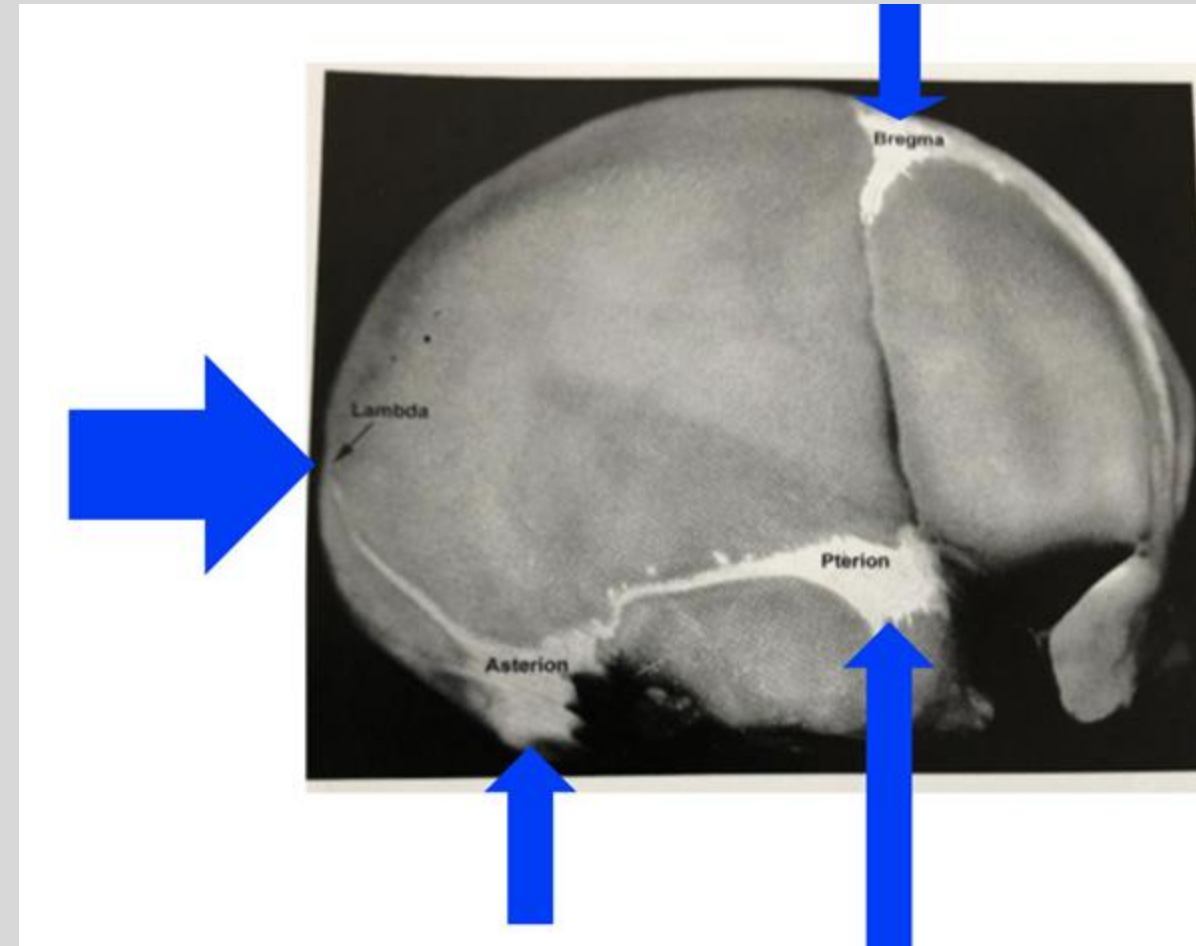


Infant Cranium

Six Fontanelles of the Infant

- Cranium in which the parietal bone is floating
- **29** cranial bones
 - 14 facial \
 - 8 neurocranium / = **22**
 - 6 ossicles
 - 1 hyoid
- **7** bones of the orbit:
- Frontal, zygomatic, maxillary, sphenoid, ethmoid, palatine, lacrimal
- Cranial vault is developed from membrane
- Cranial base is developed from cartilage
- **143 craniofacial articulations**
- Teeth are not bone, but enamel is ectoderm

II, A. F. D., & Agur, A. M. R. (n.d.). Grant's Atlas of Anatomy, 14e. In *Grant's Atlas of Anatomy, 14e*, Chapter 7: Head. essay, Lippincott, Williams & Wilkins. LINK: <https://meded-lwwhealthlibrary-com.arktos.nyit.edu/content.aspx?sectionid=115777743&bookid=1737&resultClick=1#115777750>.



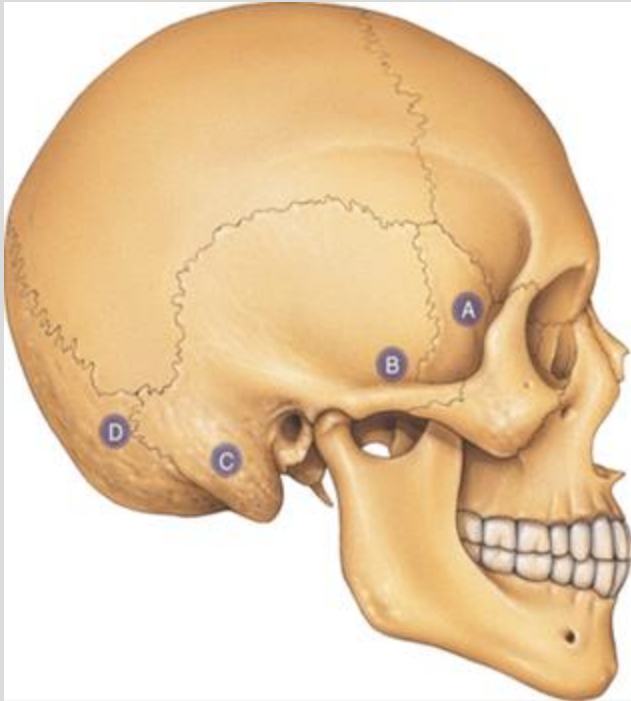
Assess, “Dur-assess” and Reassess

- Note similar to other body parts, that for all the cranial and facial techniques, one must assess for somatic dysfunction (i.e. TART- such as tenderness, soft tissue edema, muscle hypertonicity of the face or neck, restricted Frontal or Zygomatic bone motion, etc.)
- During OMT or OCF one can dur-assess and become more proficient at establishing whether this treatment is the best avenue to accomplish the intended result. (or should we modify? Try a different approach? Would direct or indirect treatment work best in this patient today? Should I go lighter or add a little more force? Etc.)
- After OMT, one should reassess by palpating the treated regions or **re-performing the technique** to see if the treated structures feel freer? Did the treatment seem to make a change in the degree of SD? In practice, we can also ask how a patient feels (subjective improvement may not be immediate, however). One can reassess by rechecking for changes in TART.

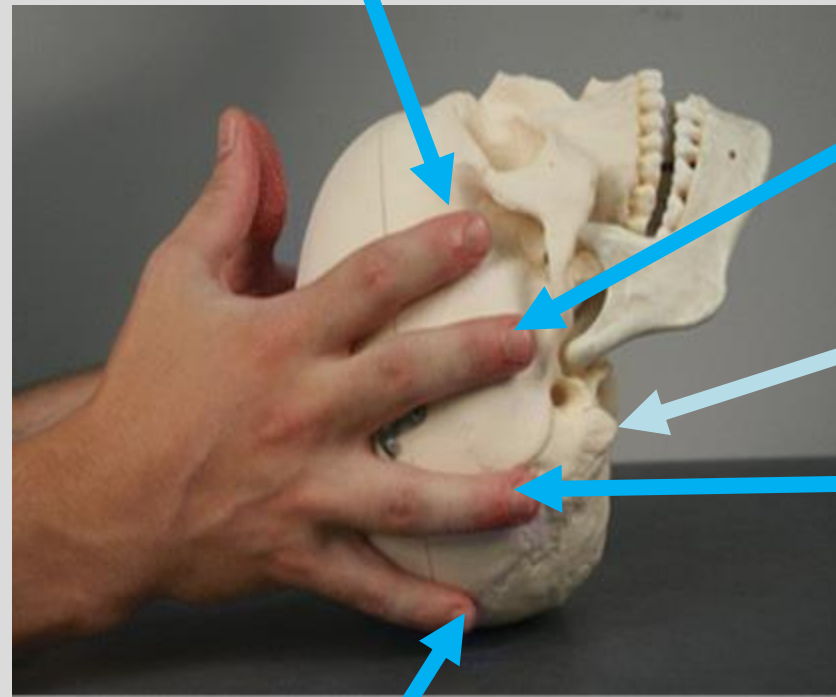
Vault Contact (a.k.a. Vault Hold)

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greater wings of the sphenoid



temporal bone
anterior to the ear

mastoid process

temporal bone
posterior to the ear

squamous portion of the occiput

Review of Physiologic Strain Patterns

Pattern	Number and Direction of Axis	Movement of Occiput & Sphenoid relative to each other	Physiologic or non-physiologic Pattern	Description of air hands from vault contact
Flexion	2 transverse axes	Move around transverse axes in opposite directions	Physiologic	"Swelling" The head expands into the operator's hands laterally and shortens vertically, so palms move laterally, fingers spread & Fingers move towards the feet
Extension	2 transverse axes	Move around transverse axes in opposite directions	Physiologic	"Receding" Opposite as above: Palms come closer to one another and cephalad as fingers approximate
Torsions (R or L)	1 A/P axis	Move around A/P axis in opposite directions	Physiologic	Hands rotate (twist) in opposite directions (one twists sup-posterior, one inf-anterior) Named for side of high GW.
Sidebending & Rotation (R or L)	2 parallel vertical axes, 1 A/P axis	Sidebending: Move in opposite directions around parallel vertical axes Rotation: Move in same direction around A/P axis	Physiologic	The hand on the side of convexity moves caudally with fingers spreading apart. The hand on side of concavity moves cephalad with fingers coming together. Named for side of fullness (convexity).

Review of Non-Physiologic Strain Patterns

Pattern	Number and Direction of Axis	Movement of Occiput & Sphenoid relative to each other	Physiologic or non-physiologic Pattern	Description of air hands
Lateral Strain (R or L)	2 parallel vertical axes	Move around vertical axes in same direction	Non-physiologic	Hands slant. Index fingers on GW of sphenoid move opp to direction of pinkies on occiput. BUT <u>Named for direction of basisphenoid</u> (which is the same direction as the pinkies, opp the direction of the GW)
Superior Vertical Strain/ Shear	2 transverse axes	Move around transverse axes in same direction	Non-physiologic	Fingers point down towards feet (wrists ulnar-deviate). <u>Named for direction of basisphenoid</u>
Inferior Vertical Shear/ Strain	2 transverse axes	Move around transverse axes in same direction	Non-physiologic	Fingers point up toward head. (wrists radial-deviate) <u>Named for direction of basisphenoid</u>
Sphenobasilar Synchronosis Compression	No axes	N/A	Non-physiologic	Feels rock-hard and heavy like a bowling ball head

Flexion



Extension



Right SB & Rot



Left SB & Rot



Right Torsion



Left Torsion



**Right Lat.
Strain**



**Left Lat.
Strain**



**Superior
Shear**



Inferior Shear



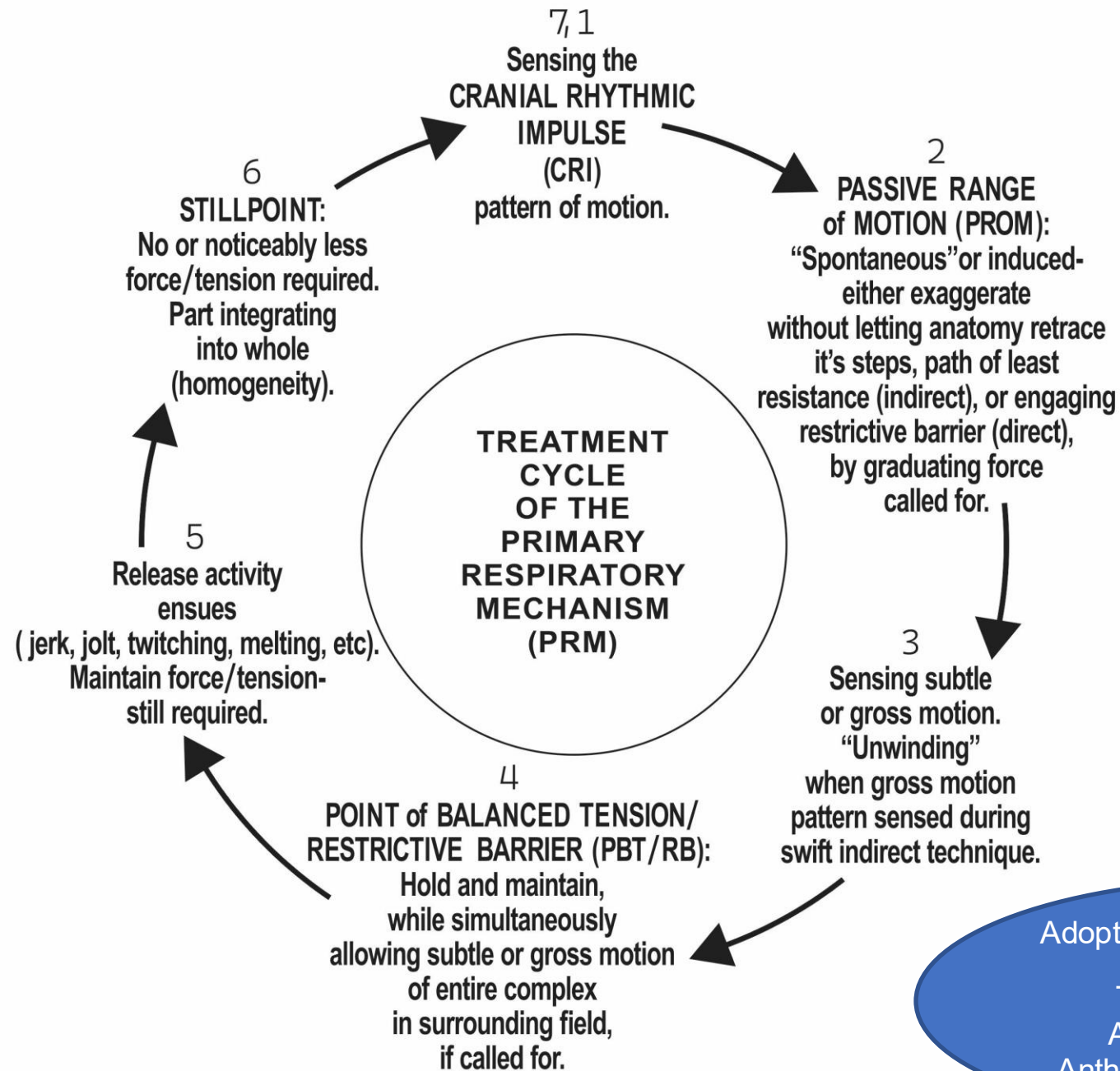
Motion Testing for Strain Patterns

- While in the vault contact, palpate the cranial rhythmic impulse (CRI) as the head moves into flexion and extension.
- During a flexion/inhalation phase, very gently (and by gently, we mean to just think about performing the motion rather than actually doing it) guide the head into the strain pattern you would like to test for. i.e. Right torsion. See how the head responds. Then relax your intention and allow the head to go through a full phase of flexion and extension. As a 2nd flexion phase begins, test for the opposite direction. i.e. Left torsion.
- Compare sides.

Motion Testing for Strain Patterns

(continued)

- Repeat for all the strain patterns.
- The direction that has more motion (the freedom) is the one the dysfunction(s) is/ are named for.
- Remember to be gentle with motion testing. You do not want to induce a strain pattern.



Adopted from The Cranial Letter
May 2000, Vol 53, Number 2
The Treatment Cycle:
Anatomy of a Release
Anthony D. Capobianco, D.O.

Treating cranial strains with two kinds of Balanced Membranous Tension (BMT)

1. **Direct BMT** – Gently guide the strain pattern present into its barriers, i.e. opposite to the way that it “wants to go.” Hold point of balanced tension (POBT) until you feel a release(s).
 - Commonly used in infants and children before sutures fully develop
 - Can be used when indicated for trauma
 - Can be used for articular restrictions
 2. **Indirect BMT** – Gently guide the strain pattern into its freedoms, i.e. the way it “wants to go.” Hold point of balanced tension (POBT) until you feel a release(s).
- Exaggeration and Disengagement are two other technique classifications that you could read about but will not be discussed in lab, nor will you be held responsible for them at this time.

V-Spread of the O-M suture with **Directing the tide technique**

V-spread is a combination of disengagement and directing the tide.

- The physician places two fingers of one hand on either side of the suture to be released and exerts gentle spread/traction to disengage the suture.
- Simultaneously, one or two fingers are placed on the point of greatest distance from the suture on the contralateral side. The intention to direct the fluid with these fingers, combined with gentle pressure, if necessary, will send a fluid fluctuation toward that suture.
- This fluid fluctuation should be palpated by the V-spread fingers and will continue to manifest an effect between the two guiding and directing hands.
- The physician adjusts the directing fingers until a response is palpated between the two V-spread fingers.
- Release or an apparent softening of the tissues will occur.



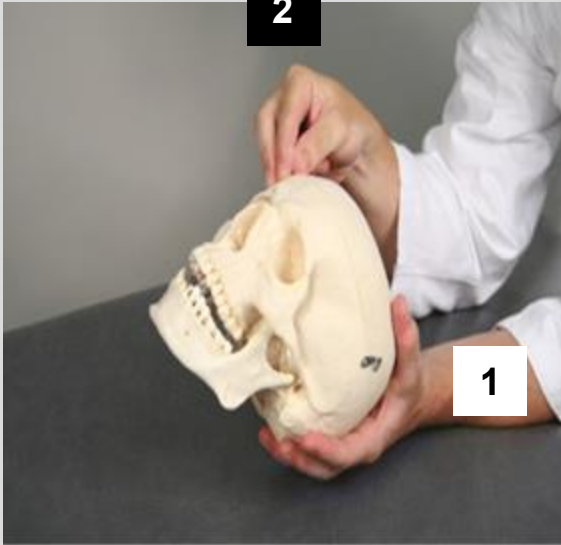
<https://osteopathicmedicine.lwwhealthlibrary.com/content.aspx?sectionid=252637062&bookid=3202>

V-Spread of the O-M Suture

(Directing the Fluid Tide Technique)

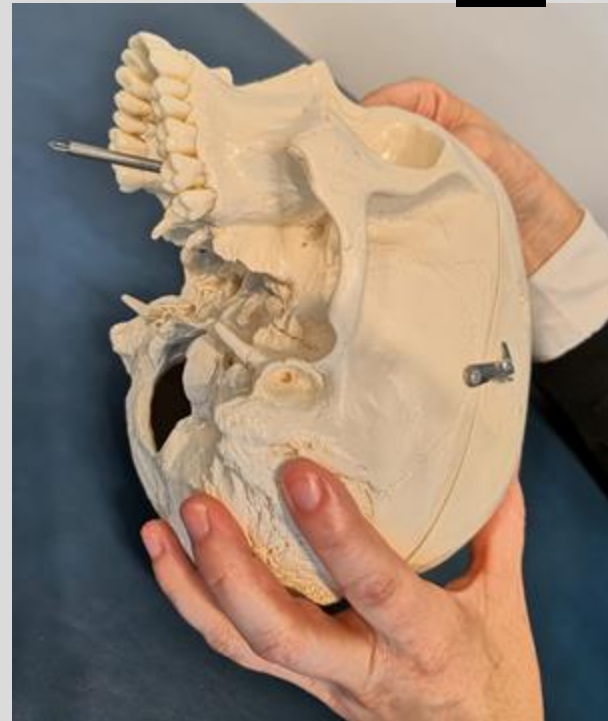
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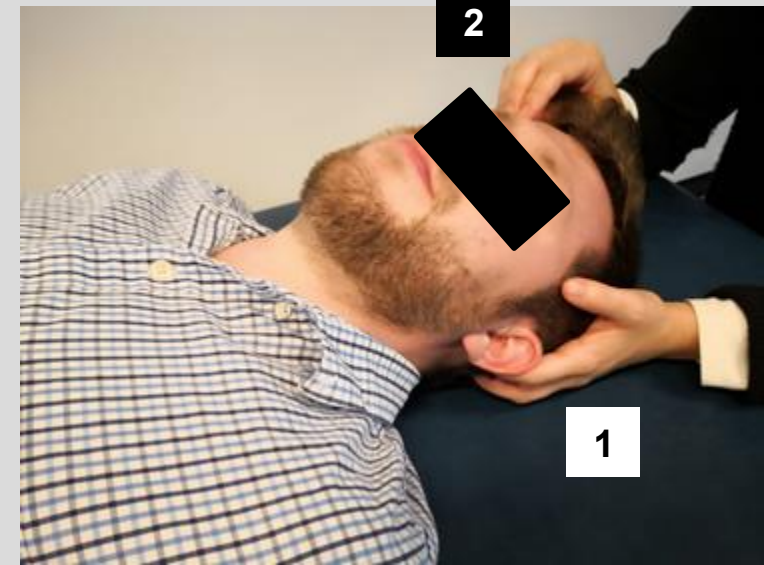


1 = first hand, index and middle finger on either side of restricted suture

2 = second hand, opp side of head at furthest point from restricted suture



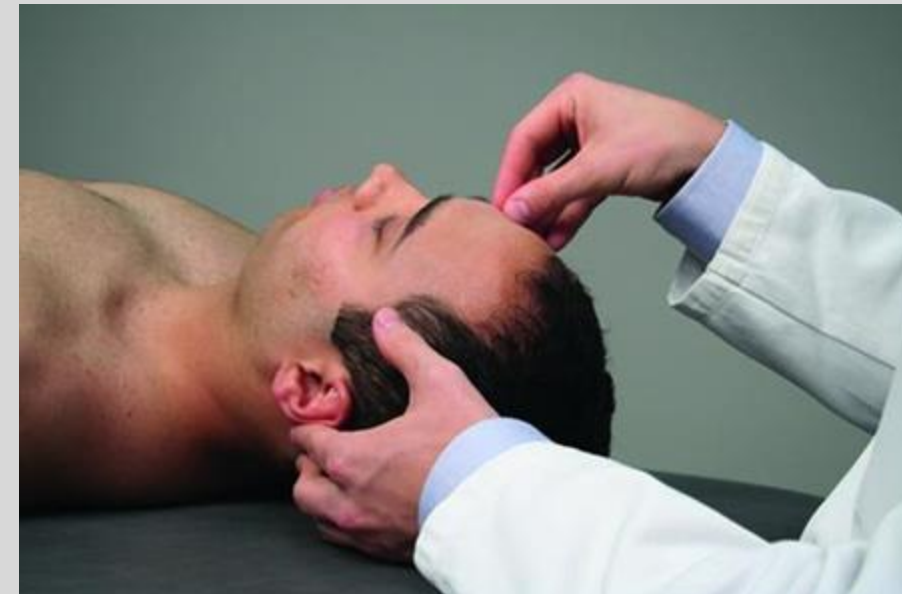
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V-Spread & Directing Tide continued (Nicholas-Sutural Spread)

1. The patient lies supine, and the physician is seated at the head of the table with both elbows resting on the table, establishing a fulcrum.
2. The physician places the index and middle fingers on the two sides of the patient's restricted suture.
3. The physician places one or two fingers of the other hand on the patient's cranium at a point opposite the suture to be released.
4. With the lightest force possible, the physician directs an impulse toward the restricted suture with the hand (on the forehead) opposite the suture, initiating a fluid wave. The object is not to physically push fluid through to the opposite side. Instead, the physician is using the fluctuation of the cerebrospinal fluid to release the restriction. The physician uses intention to initiate this wave; this method contracts the fewest muscle fibers and so applies the slightest force.
5. This fluid wave may bounce off the restricted suture and return to the initiating hand, which should receive and redirect the returned wave toward the restricted suture.
6. This back-and-forth action may be repeated for several cycles before the physician feels the suture spread and the wave penetrating the suture does not return to the initiating hand.
7. The rate and amplitude of the CRI at that suture are retested to assess the effectiveness of the technique.

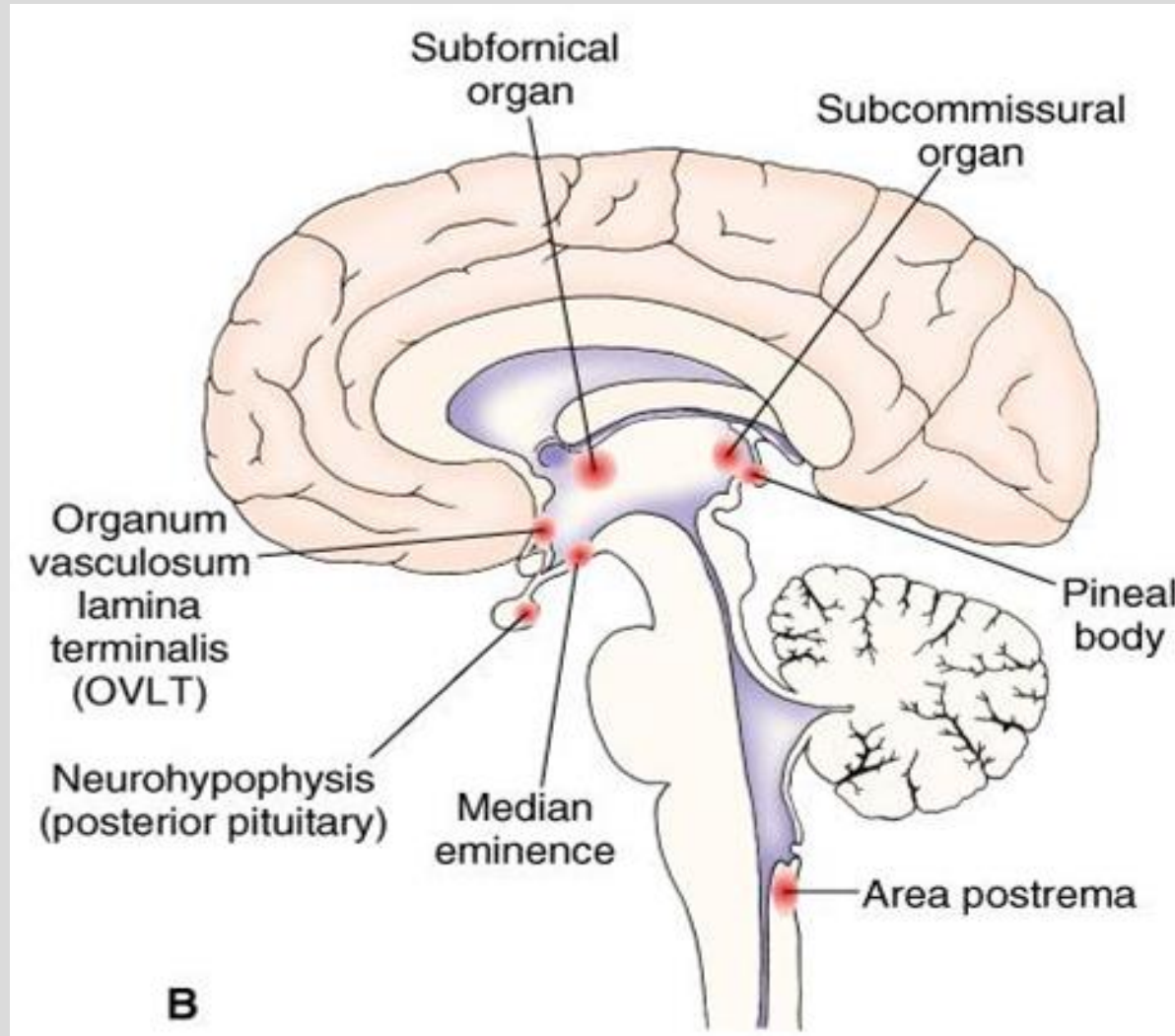


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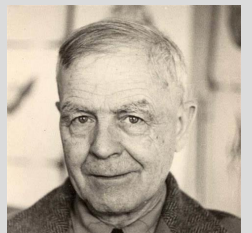
CV4 & the Circumventricular Organs

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- CV4 is considered one of the most useful cranial techniques
- Augments the healing capabilities of the patient, relaxes the patient, and improves the motion of the CRI
- Can be used to relieve headaches, reduce fever, assist in difficult/stalled labor, relieve congested sinuses and lungs, reduce edema, and reduce the effects of trauma (e.g. whiplash injury)
- May also assist in the resolution of secondary somatic dysfunctions throughout the body.



CV4

1. The patient lies supine, and the physician is seated at the head of the table with both forearms resting on the table, establishing a fulcrum.
2. The physician crosses or interlaces the fingers of both hands, cradling the patient's occipital squama.
3. The physician places the thenar eminences posteromedial to the patient's occipitomastoid sutures. *If the thenar eminences are on the mastoid processes of the temporal bones, the compression that follows will bilaterally externally rotate the temporal bones, which may cause extreme untoward reactions (Figs. 18.20 to 18.23).*
4. The physician encourages extension (medial springing to create narrowing) of the patient's occiput by following the occiput as it moves into extension.
5. The physician resists flexion by holding the patient's occiput in extension with bilateral medial forces. *Note:* The occiput is not forced into extension. Rather, it is prevented from moving into flexion. It is as if the physician is taking up the slack created by extension and holding it there.
6. This force is maintained until the amplitude of the CRI decreases, a still point is reached, and/or a sense of release is felt (a sense of softening and warmth, a bead of sweat on the brow or a change in respiration).
7. As the CRI resumes, the physician slowly releases, allowing the CRI to undergo newfound excursion.
8. The rate and amplitude of the CRI are retested to assess the effect of the technique.

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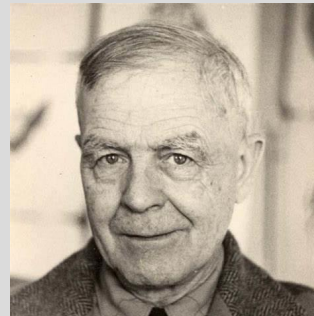
CV4: Compression of the Fourth Ventricle

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- With his years of research, practice and experience, Sutherland believed that the deepest healing came from the Stillness of the Tide. “It is the Stillness of the Tide, not the stormy waves that bounce upon the shore, that has the potency, the power.”
- Dr. Sutherland observed that when the cranial-sacral mechanism was brought to a short rhythmic period of fluctuation, a stillness was revealed at the center of the tide, and a transmutative process unfolded nourishing the cells of the body.

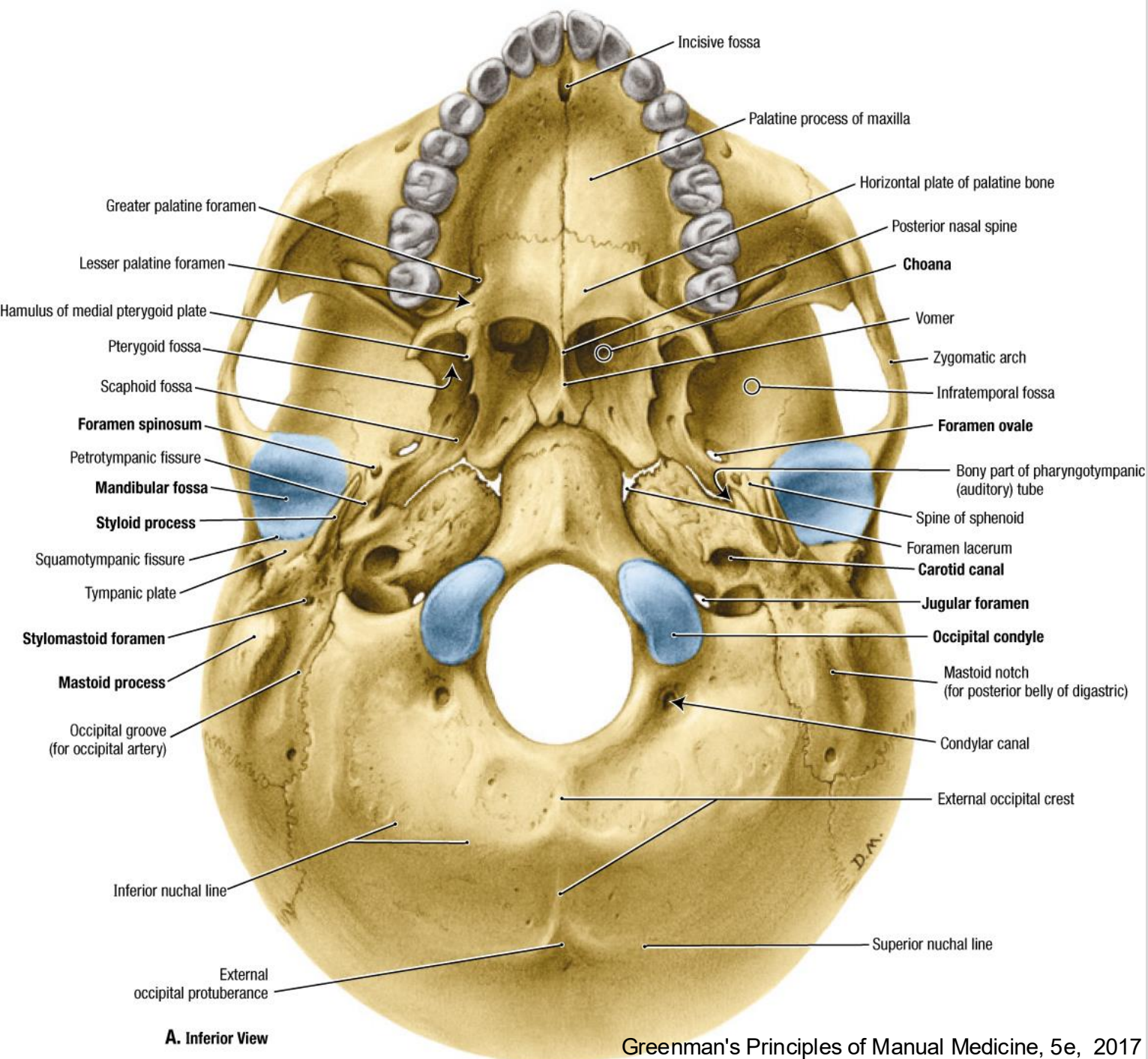


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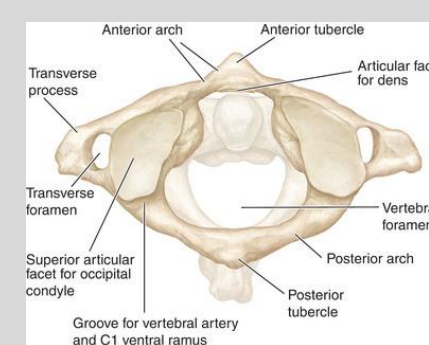
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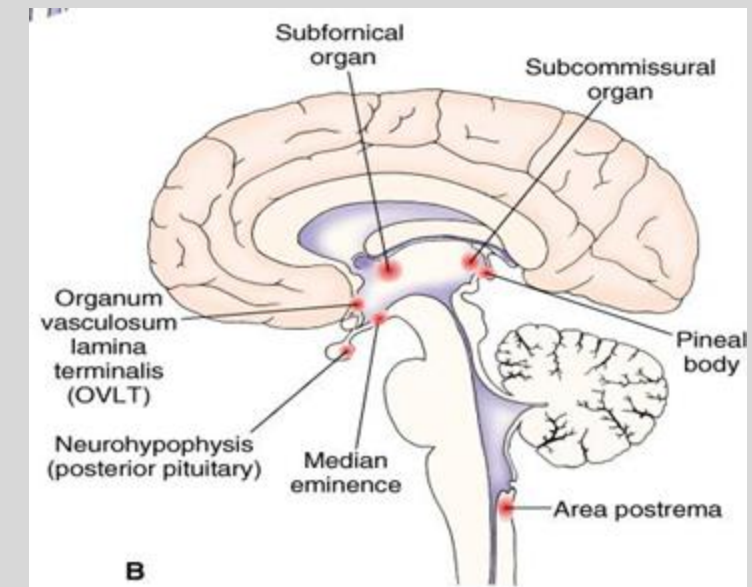
Greenman's Principles of Manual Medicine, 5e, 2017



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Grant's Atlas of Anatomy, 15e,
2021



Nicolas.
Atlas of
Osteopathic
Techniques
4th ed. Ch18

Some indications for treating patients with Osteopathy in the Cranial Field include:

- Headaches
- Mild to severe whiplash strain and sprain injuries
- Post-concussive syndromes/ TBI
- Vertigo and tinnitus
- Cranial neuropathy and/or nerve entrapment
- Trigeminal neuralgia
- Bell's palsy
- Otitis media with effusion and serous otitis media
- TMJ and malocclusion
- Sinusitis
- Colic
- Plagiocephaly/Torticollis
- Feeding difficulties
- Pathology involved with vagus (Cardio,Resp,GI), jugular vein
- Eye complaints
- Pituitary - Neuro-endocrine-immune
- Stalled Labor/ ineffective contractions
- Etc.

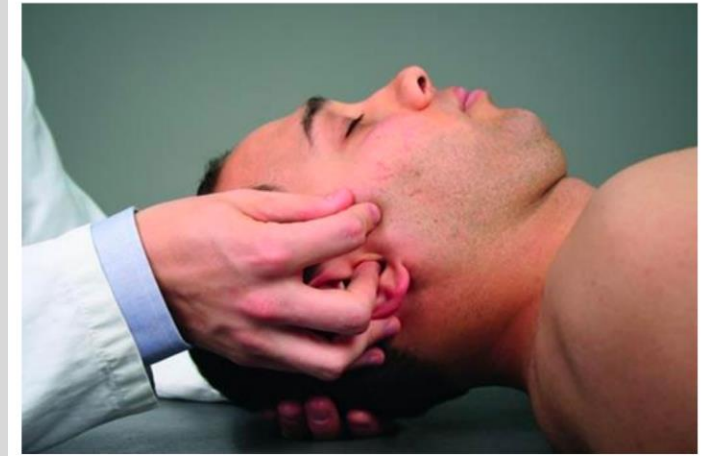
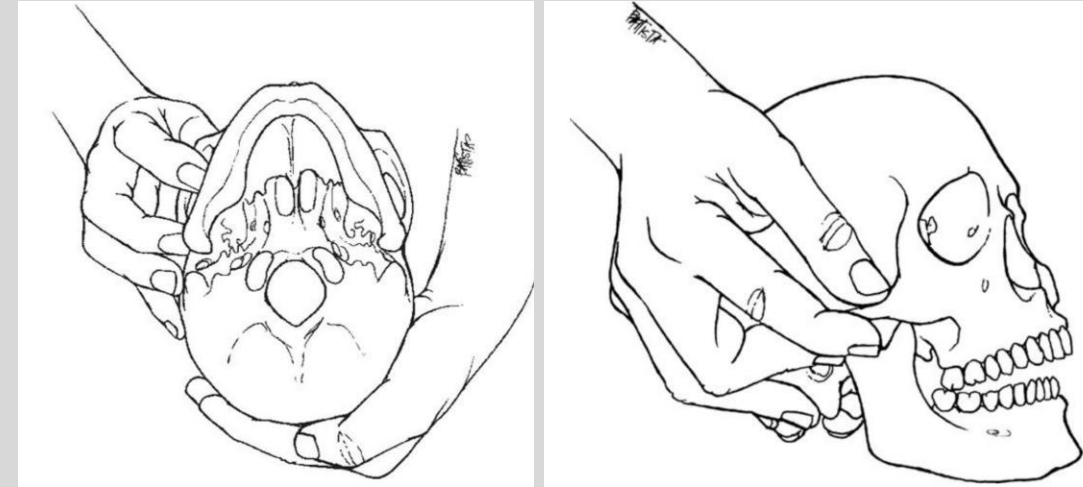
Unilateral Temporal Hold/Contact

1. The patient lies supine, and the physician is seated at the head of the table with both forearms resting on the table, establishing a fulcrum.
2. The physician's left hand cradles the patient's occiput.
3. The physician's right thumb and index finger grasp the zygomatic portion of the patient's right temporal bone, thumb cephalad, and index finger caudad.
4. The physician's right middle finger rests on the external acoustic meatus of the ear.
5. The physician's right ring and little fingers rest on the inferior portion of the patient's mastoid process

When to use this hold?

For Diagnosis of Temporal Dysfunctions.

It can also be used as a contact for treatment if one or more temporals are found to be restricted (to be learned at a future date).

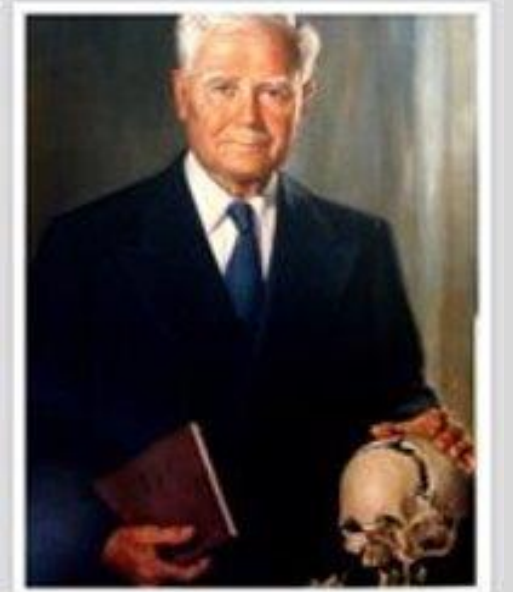


Summary Slide

1. You have located various cranial **landmarks** and reviewed and practiced sensing the **CRI and PRM utilizing cranial contacts** (holds) including a new contact: **Unilateral Temporal Hold**.
2. You will have reviewed **cranial strain patterns** and motion tested for them on your partner
3. You will have treated the strains present using **indirect and/or direct BMT** technique
4. You will have practiced **V-spread and directing the tide** to release the O-M sutures AND have attempted to apply the **CV-4** technique to your lab partner.
5. Please use your hands and mind to continue to integrate your understanding of anatomy, physiology and pathology in your understanding of health and various dis-ease states.

General References

- [Foundations of Osteopathic Medicine: Philosophy, Science, Clinical Applications, and Research, 4e - Ch 38](#)
- [Nicholas, Atlas of Osteopathic Techniques, 4e - Ch 18](#)
- [Eileen L. DiGiovanna, Christopher J. Amen, Denise K. Burns, An Osteopathic Approach to Diagnosis and Treatment, 4e, Wolters Kluwer 2021: Chapter 105: Diagnosis and Treatment of the Cranium: topics: Principles of Dx, Princ. Of Tx, BMT and V-Spread](#)
- *Magoun 3rd Ed Osteopathy in the Cranial Field
- *The Cranial Bowl by WG Sutherland, D.O. 1944



Lab Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>

